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CAM_C101_W5_Mini-project
Anomaly Detection April 2025

Engine Anomalies Throughout the Fleet

Executive Summary

A robust anomaly detection system to counter fleet anomalies is prescribed within a recommended 1-5% range. It predominantly favours Isolation Forest method.

1. Introduction

This brief report centres on anomaly detection in 19,535 engines using six features, on a pre-processed dataset. Metric values include original non-normalised records using pressure, temperature and rpm scales, indicating poor correlation. To measure anomalies, IQR, One-class SVM and Isolation Forest methods were used in order to compare results generated from techniques with different strengths and weaknesses. The purpose of this work is to describe changes that would be made in order to attain the recommended 1-5% outlier values across the fleet.

2. Exploratory Data Analysis (EDA)

Pre-processing and cleaning verified that the data did not contain null values, missing values and duplicate values. Mean and median values are reported to show the range of values. Visualizations that plot these core metrics and an approximation of the Gaussian distribution for the dataset are provided at the end of this short report (A.1). Shapiro-Wilk tests confirmed that features were not normally distributed too.

A correlation heatmap indicated moderate relationships between temperature and pressure features, suggesting some dependency worth accounting for in multivariate models.

Feature	Mean	Median
Engine rpm	791.24	746.00
Lub oil pressure	3.30	3.16
Fuel pressure	6.66	6.20
Coolant pressure	2.34	2.17
Lub oil temp	77.64	76.82
Coolant temp	78.43	8.35

3. Statistical Anomaly Detection (IQR)

The Interquartile Range (IQR) is a statistical method intended to capture the normal range of a given dataset feature. Such bounds are highly variable on a non-normalised or differing scale of values. However, it was possible to ascertain that 977 values are extreme and exceed the ninety-fifth percentile; fuel pressure and lubrication oil temperature are particularly high with a total 5,124 outliers detected (A2). Engines with a minimum threshold of two outliers from the six different features were considered anomalies (n=4636, 23.7%).

Feature	Outliers
Engine rpm	464
Lub oil pressure	66
Fuel pressure	1135
Coolant pressure	785
Lub oil temp	2617
Coolant temp	2

IQR is considered to be robust to outliers with strict upper and lower bounds (.25% and .75%) independent of data distribution type. Typically, it under-performs in very large datasets and it should not be used in isolation on a dataset of these proportions with highly-skewed data (A1). Therfore, machine-learning methods One-class Support Vector Machines (SVM) and Isolation Forest (IF) were introduced to detect anomalies.

4. Machine Learning-Based Anomaly Detection

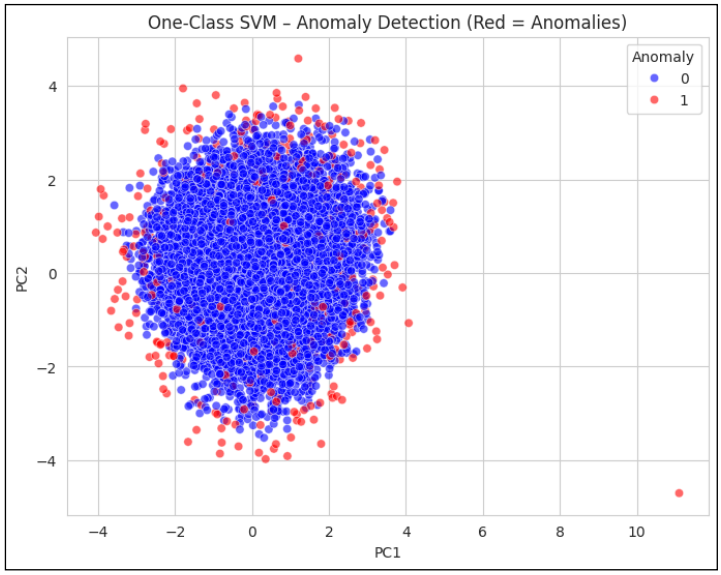
Unsupervised learning represents the class of algorithms that perform analysis on unlabeled datasets. Machine learning models One-Class SVM and IF require normalized input for realisable gains from their

5.1 One-Class SVM

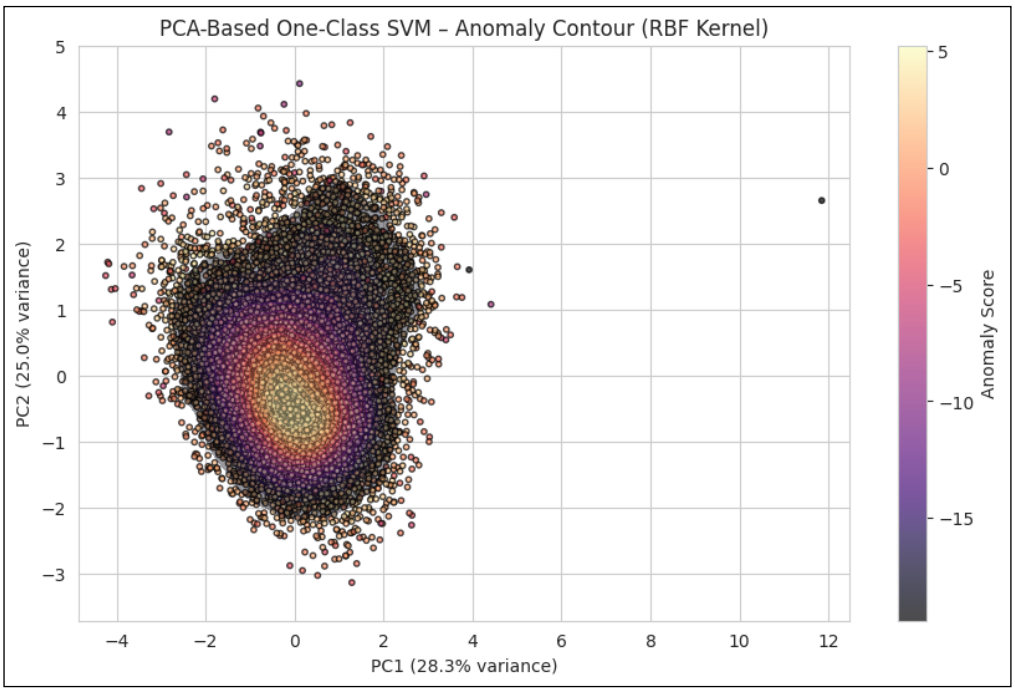
results, that is to say distance-based anomaly detection applied on a universal scale. The data was initially scaled to perform Principal Component Analysis used also for 2D visualizations and improve the performance of algorithms as follows:

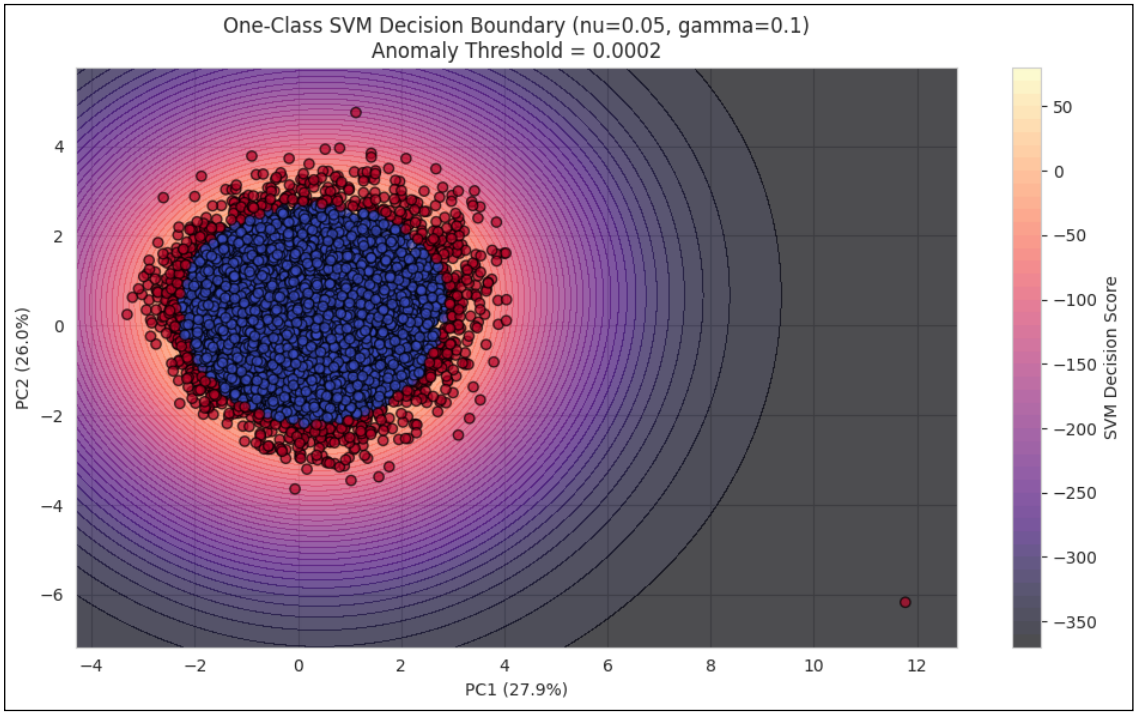
It should be noted that Engine rpm represents a significant impact over the PC1 and PC2, which is potentially a result of sprawling data points (sigma = ~267).

The 585 anomalies detected by SVM could then be identified in a visual 2D representation using PCA; outliers are represented in red to the right. One-class SVM uses a kernel density, in this case RBF, scaled by parameters nu and gamma. The use of the parameters was experimental and a nu of 0.03 was found to be optimum in experimentation with a gamma scaled accordingly.

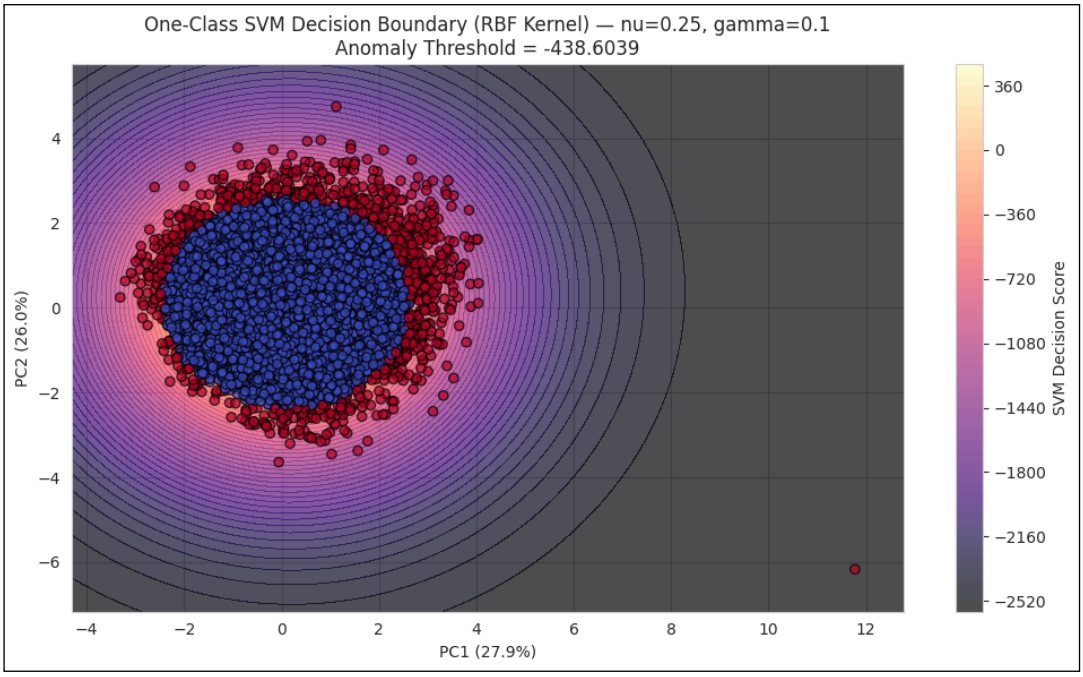


As a result of difference between the results generated by IQR, which cannot respond well to multivariate data, and SVM, the Expected Variance (EV) metric was also calculated in order to identity the extent of the variance represented by the six features after performing RFE. This process eliminated two features and the SVM model was re-calculated using only Lub oil pressure, Fuel pressure, Lub oil temp and Coolant temp. Although principal component PC1





indicates EV representative of 28.3% variability in the data, which is reasonably strong, it does not immediately support IQR results.



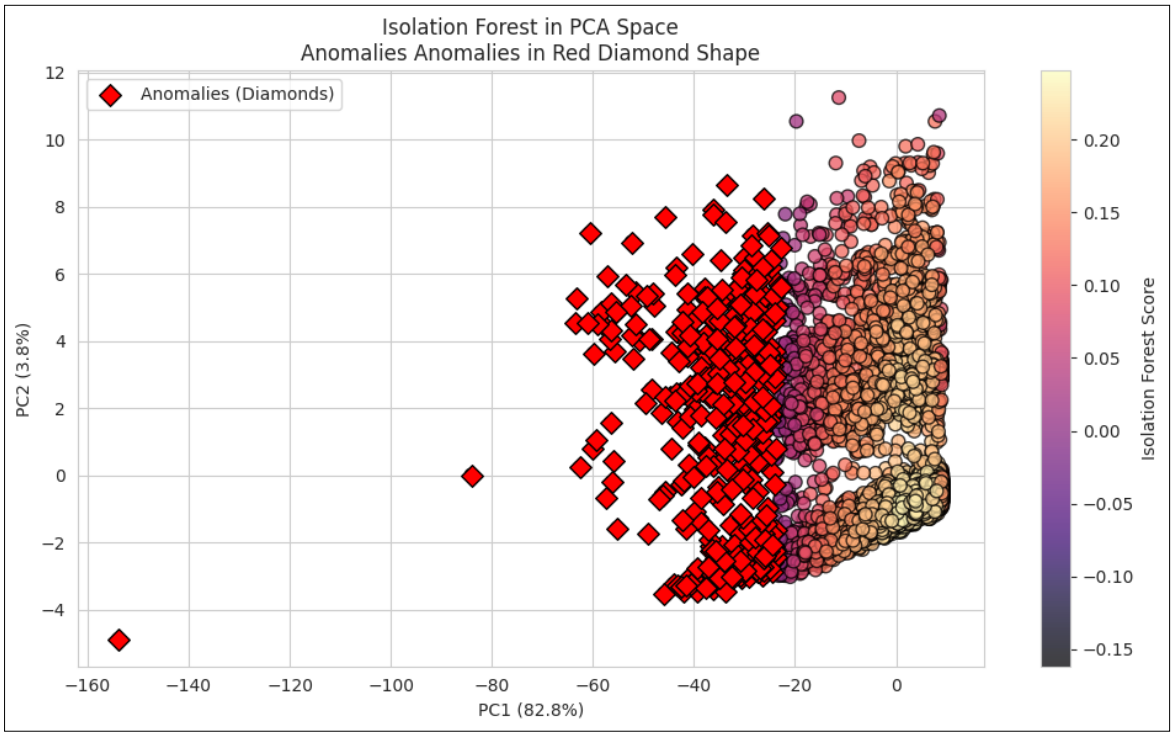
To improve the result, a second experiment was carried out. The contour lines show model boundaries at various decision points and the colour bars a data point's normality.

However, improvements to the results can be found with adjusted parameter values (kernel=rbf, nu=0.25, gamma=0.1). This analysis was supported by the use of the decision score metric and improved the PC1 score to EV=27.9% with 586 anomalies overall.

5.2 Isolation Forest

An IF model was chosen with the desired parameters drawn from estimation of the number of isolation trees in the forest and a maximum number of samples to train each estimator; this is known as the contamination rate. The contamination of a data set is equivalent to the proportion of outliers it contains. Thus, this parameter is useful to run with the default 'auto' value for a clear overview of the likely result. The randomisation of the estimator is also controlled (n_jobs = -1) by a parameter which used all available processors in parallel for both a fit and prediction score.

The scaled data was run in ten tests in order to identify the strongest results. The IF was run using a contamination of 2.5%, firmly within the desired 1-5% range and it achieves a PC1 of 82.8% for one feature with an anomaly per engine.



6. Conclusion

The IF detected 489 (2.5%) anomalies with contamination set to 2.5%. This figure is significantly less than the IQR detected values (anomalies=5,124) and SVM (anomalies=585). The clarification of the result provided by IF is particularly clear in EV=82.8% where PC1 explains the majority of the variance.

7. Recommendations

The IF is the preferred anomaly detection method, a stable and interpretable method. The threshold has balanced overfitting from false positives with anomaly detection. The consistently low coolant or lubrication pressure has adversely affected the engines.The IQR method offers a strong baseline, while Isolation Forest provides a scalable and interpretable enhancement.